

What Drove Success at the 15th Video Browser Showdown? A Comprehensive Interaction-Logging Analysis

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Abstract

In 2026, the Multimedia Modeling conference in Prague hosted the fifteenth edition of the Video Browser Showdown (VBS) competition. Yet, for the first time, two participating systems implemented full-scale interaction logging frameworks, enabling a detailed analysis of the search process beyond traditional score-based evaluation. In this paper, both systems are introduced and described with a focus on their user interactions. To enable compact presentation and analysis of logs, all interaction types are further grouped into more abstract events, forming an interaction taxonomy hierarchy. Finally, we reveal the applied search strategies and analyze which factors drive success and failure. The results reveal a clear dominance of iterative, high-frequency text query reformulation with result set inspection, leveraging the power of modern CLIP-based models across most competition categories. In around 10% of cases, users also relied on advanced system features to achieve good performance at VBS, mostly on challenging homogeneous datasets.

CCS Concepts

• **Information systems** → **Multimedia databases; Retrieval models and ranking.**

Keywords

Interactive Video Search, Logging Analysis, Text-image Search

ACM Reference Format:

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1 Introduction

Efficiently searching in large video collections requires tailored retrieval systems to meet user requirements. Such systems typically offer a variety of query options that users can leverage to express their search intent. While advances in machine learning models continuously improve retrieval effectiveness, human interaction remains essential: users must validate results and, in most cases, iteratively refine queries to ensure that retrieved results match their intended goal [41]. Thus, an interactive search process is created, in which users continuously alternate between issuing queries, inspecting result lists, and adjusting their strategy based on partial findings. This raises an important question: how can we evaluate retrieval systems that reflect interactive user search?

A widely used approach is to evaluate systems in competitive benchmarks designed to reflect interactive, time-critical search. The Video Browser Showdown (VBS) [33] is a prominent example in which research teams develop retrieval tools and apply them to a variety of retrieval tasks across challenging video collections [21, 28, 39]. Over the years, VBS has demonstrated steady progress: more tasks can be solved than ever before, and semantic text-based retrieval, frequently powered by embedding models such as CLIP [22], has become one of the dominant search strategies. At the same time, the competition setting primarily captures outcome-level performance (e.g., task completion and time), while offering only limited support for explaining why performance differs across teams or users. Addressing this issue, several studies have collected interaction logs to analyze user behavior [12, 40]. However, these logs typically record only abstract action types (e.g., text queries or relevance feedback), limited metadata (e.g., user identifiers and timestamps), and the resulting ranking. This level of abstraction makes it difficult to reconstruct which strategies users actually applied during a session, thereby preventing the disentanglement of performance gains driven by system capabilities from those driven



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by user interaction and strategy. Broadly, the interactive search process remains only weakly observable at the user level, limiting our ability to analyze how users adopt and combine interaction mechanisms, or how preferences influence systems in practice.

In this work, we take a step toward making interactive video retrieval more explainable by enabling a detailed analysis of the user search process. To this end, we implemented fine-grained interaction logging in Exquisitor and PraK, which participated in VBS 2026 and achieved 3rd and 1st place, respectively. Our contributions are threefold: (1) we introduce fine-grained logging for both systems, enabling detailed reconstruction of user search behavior; (2) we propose a interaction hierarchy that maps system-specific actions to high-level categories, enabling unified visualization and reporting of interaction logs; and (3) we use these logs to analyze user behavior and success factors at VBS 2026, revealing the central role of iterative text querying and result inspection, while advanced query features were used almost exclusively on homogeneous datasets.

2 Related Work and Background

Interactive Video Retrieval Competitions. Competitions such as VBS [12, 33, 40], the Lifelog Search Challenge (LSC) [36, 38], and CASTLE [24], require participants to develop systems that enable efficient search in large, challenging video collections. These competitions typically have two phases: In the first phase, researchers prepare systems and preprocess the video collections to enable *interactive* use during the competitions. Video collections can vary substantially in both content and scale. VBS 2026 therefore employed three datasets: (1) V3C [28], comprising more than 3,800 hours of everyday-life footage; (2) the highly homogeneous VBS-LHE [21] collection of laparoscopy videos, containing more than 100 hours of material; and (3) the underwater MVK dataset [39], with approximately 28 hours of video. While some include metadata, such as timestamps indicating when they were created or simple textual descriptions, most are unannotated, simulating real scenarios. The second phase is the competition itself, which happens live and in person: A range of tasks is presented to participants, who must solve them interactively using developed systems under time pressure. Some competitions also include novice sessions, where participants unfamiliar with the system attempt to solve tasks to evaluate general usability. The most common task types are:

- *Known-Item Search (KIS):* KIS requires users to find a single section in a video. This section is either previously shown to them (Visual KIS) or specified by a textual description (Textual KIS). In this year’s VBS, conversational KIS tasks were also evaluated, allowing participants to ask live judges for text hints to improve their search results.
- *Ad-Hoc Video Search (AVS):* For this task, users need to find *all* matching moments in videos that fit a description.
- *Question Answering (QA):* In QA tasks, users need to find a correct (textual) answer to a given question. Most often, the answer can be found within a single video. Consequently, users must first identify the video using a hint, then search within it for the essential information.

For specific tasks employed in VBS 2026 and previous editions, we refer readers to the VBS-Repository [23].

Video Retrieval Systems in Controlled Benchmarks. The last decade of video retrieval competitions revealed countless systems with highly specialized functionalities. For a detailed overview and history of approaches, we refer to surveys [12, 33, 40] and individual publications of retrieval systems [7, 20], and limit ourselves to describing recent trends and commonalities. Performant retrieval systems combine a preprocessed database with frontends that enable interactive querying and browsing. Usual preprocessing pipelines include shot detection, keyframe extraction, and, finally, a multi-modal embedding model (such as CLIP [22]) that condenses raw video collections into storage- and compute-efficient vectors. During the competitions, vast video collections can thus be efficiently queried by text within a few seconds. These text queries produce a ranked list of candidates (either keyframes or short video sequences), which often serve as the initial step in a complex search path for users. The search path involves browsing phases, in which users inspect the ranked list for target(s) or query the collections again, either to refine the results or produce novel result sets.

Evaluation of User Behavior in Information Retrieval Systems. User-centric evaluation of retrieval systems is an active area of research in interactive information retrieval. Existing work studies both isolated search behaviors, such as inspecting ranked image grids [10, 19] and query reformulation [2, 6], and broader behavioral patterns that unfold across entire search sessions [3, 5, 9, 11]. At the same time, dedicated logging frameworks have been developed to support the scalable collection of interaction data [1].

In VBS, evaluation is standardized through the Distributed Retrieval Evaluation Server (DRES) [27, 29]. For each submission, it records the submitted answer, such as a text response or video moment, together with the submission time, status, and user. However, these records primarily capture the outcome of a search session. As a result, although user behavior can substantially affect a system’s score, the interaction path leading to a correct answer is neither captured in detail nor used for system comparison.

Several VBS studies therefore complement final scores with analyses of interaction logs that capture users’ queries, returned result lists, and parts of the browsing process [4, 12, 13, 17, 26, 40]. These logs provide insight into how systems are used in practice, for example, by revealing which query modalities dominate (e.g., text, OCR/ASR, or visual similarity search), how frequently users reformulate queries, and how early relevant results appear in a session. While logging formats and coverage vary across participating systems, post-hoc preprocessing and normalization allow researchers to derive comparable indicators for success, such as best observed ranks of relevant items and the delay between retrieving and submitting a correct candidate. While efforts were made to track users’ exact search paths, technical problems, incompleteness, and the lack of standardized logging prevent a more detailed analysis. Schall et al. [30] extend the official VBS 2022 evaluation by analyzing how different interaction patterns of systems contribute to success. Their findings underline that competitive performance is not solely a function of retrieval quality, but also of how effectively users can exploit advanced search mechanisms. Another case study presented by Lokoč et al. [16] contrasts automated text-to-video retrieval with interactive search workflows, illustrating how ranking models can serve as building blocks for longer interactive search

paths. In [14, 15], authors presented a log analysis of the VIRET tool performance at VBS 2019, linking result logs to submission times. The paper also introduces query modification statistics in time for different query types and a list of utilized multi-modal temporal queries. Finally, Rossetto et al. [25] further show that large-scale remote evaluations using DRES improve statistical reliability and reproducibility of competitive video retrieval benchmarks while highlighting the importance and current limitations of standardized interaction logging.

In contrast to prior studies that rely on incomplete logs and performance evaluations, our analysis is based on complete interaction logging, enabling reconstruction and explanation of the full user search process to propose improvements.

3 Compared Exquisitor and PraK Systems

This section presents an overview of two VBS systems with a focus on the set of supported interactions depicted in Figures 1 and 2.

3.1 Exquisitor

Exquisitor is a multimedia analytics system designed to support large-scale search and exploration tasks achieved through collaborative search approaches between text search and relevance feedback [34]. Relevance feedback follows a human-in-the-loop paradigm, allowing users to label retrieved items as positive (relevant) or negative (irrelevant) based on their task. The system then iteratively updates its internal model (a linear Support Vector Machine) to better align with the user’s intent in their search path. Furthermore, the text search enables users to quickly identify strong positive examples by leveraging natural language queries and metadata filtering. Relevance feedback further enables quick navigation of the collection and helps users gain insight into what to search for or refine in subsequent text queries.

Exquisitor operates on keyframes that represent either a moment (specific time) or a segment (start and end time) in a video. The SigLIP [42] model is then applied to convert the extracted keyframe images to embeddings.

In the context of the Video Browser Showdown, Exquisitor has been a participant for several years, starting as a system oriented towards a slow, interactive learning process, and has since evolved into a fast-paced collaborative search platform with text search and relevance feedback. The interface of the system supports collaboration of these search approaches by enabling users to track and interact with their searches and search results within dedicated session tabs. Within a tab, users can label items as positive and negative from any search result to construct a relevance feedback model that can perform feedback-driven search on the collection. The results of a feedback search can also be used to further label positive and negative items to adjust the model. The user can inspect and re-adjust the positive and negative items of their feedback model. Aside from these core search strategies, Exquisitor also supports temporal search, by letting the user select text queries from the tabs search list, and through a dialog set the temporal order (by default it uses the selection order), and perform the search.

The current Exquisitor system provides 58 interactive elements that can alter the visible system state, where Figure 1 depicts a hierarchical overview of the interactive elements. For simplicity,

only event interactions captured during the VBS competition are shown, and not all 58 types. For analytical clarity across the two systems, we categorize these interactions into five classes at the top level of the hierarchy:

- **Formulation interactions** support the construction, refinement, or modification of queries prior to execution. In Exquisitor, these include composing and editing textual queries, selecting and adjusting metadata filters, labeling result or video summary items as positive or negative for relevance feedback queries, selecting and ordering of text queries for temporal queries, or revising the current labeled items of the feedback model. These interactions define the search intent but do not trigger the retrieval.
- **Query Submission interactions** initiate the retrieval process by submitting a query to the backend. This typically occurs through activating a search control (e.g., clicking a search button or confirming input via *Enter* key). The system then processes the request and returns the 50 top-ranked results, which are presented in the interface’s grid.
- **Traversing interactions** where the user navigates through different sets of search results.
- **Browsing interactions** are dedicated to the exploration and inspection of retrieved results. In Exquisitor, users can switch between grid-based and row-based result views, click on an items keyframe to access the corresponding video in a side panel, operate the integrated video player, and navigate the video summary (grid of keyframes below video player).
- **Other interactions** that affect the system session context but do not have an influence on the active search process. Examples include selecting a dataset, competition events (i.e., submissions), and organizing or renaming tabs. In the hierarchy, only submission-related events are highlighted for space efficiency.

3.2 PraK

PraK was first developed for VBS 2024 [18], building on the vast experience of successful systems from previous competitions [4, 17]. The key design feature of the system was a state-less video data service component that provides abstraction of various types of datasets and their specifications. Internally, the component represents videos as sets of selected keyframes with preprocessed embedding vectors using CLIP [22, 31]. The similarity search operation is a dot product between a query vector and the stored vectors, efficiently implemented as a simple matmul operation. Based on access to general retrieval services (e.g., from a selected dataset, get top 1000 items related to a given text or image query), developers can design various front-end/back-end applications and interactive search approaches, potentially maintaining the state of iterative processes. For a technical description of the architecture, we refer the reader to Lokoč et al. [18].

The current version of PraK offers a rich set of interactions, as depicted in Figure 2. All of the presented interactions affect the state of the system or the search process. In the top layer of the tree, the interactions are divided into the five high level classes:

- **Formulation interactions** are comparable to Exquisitor. A difference is that PraK implements a sketching tool that

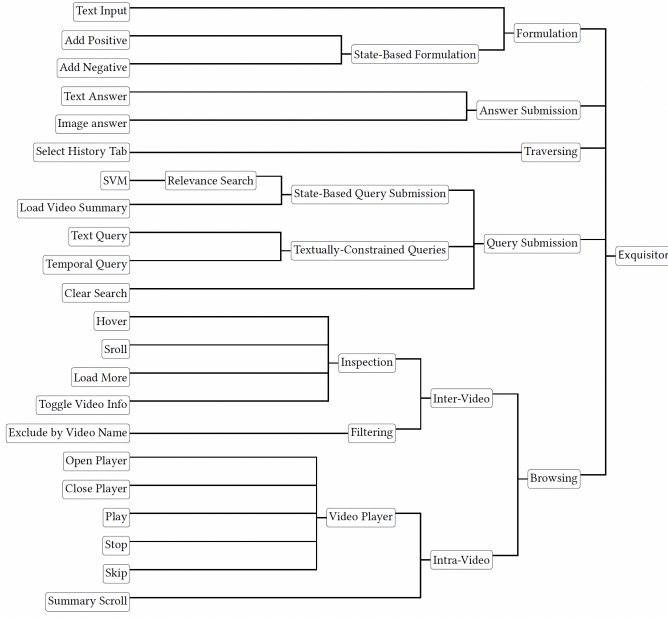


Figure 1: Exquisitor interaction hierarchy

allows users to localize their textual queries [8]. Furthermore, PraK does not support metadata specifications.

- **Query Submission interactions** are equally defined as in Exquisitor. PraK comprises text queries with optional static, and dynamic localization, temporal queries, image queries, Bayes/Rocchio relevance feedback iterations, video summary, and reset actions.
- **Traversing interactions** allow users to undo/redo certain actions, such as query submission and filtering. Traversing is implemented linearly using forward and backward buttons.
- **Browsing interactions** represent the class of interactions that allow users to inspect the returned candidate sets. Filters and re-arrange result grids, image detail inspections, or classical video interactions. We note that these events are often frequent (e.g., scrolling, or mouse actions) and require specific aggregations for effective visualization.
- **Other interactions** represent the remaining events such as interface control and submission actions.

In summary, both Exquisitor and PraK offer a non-trivial set of interaction events that may lead to various result sets, their inspection and further refinement. Together, they provide clear evidence of the search process that can be recovered once all events are logged, providing a take away for the next VBS competition. We also emphasize that the proposed hierarchy of more abstract events allows for unified visualization and reporting, although the underlying implementation details differ.

4 Interaction Log Analysis

This section provides an analysis of interaction logs collected from six expert users at VBS 2026. All of them were directly involved in the development of the respective systems. For comparing logs across systems, we use the hierarchies specified in Figure 1 and

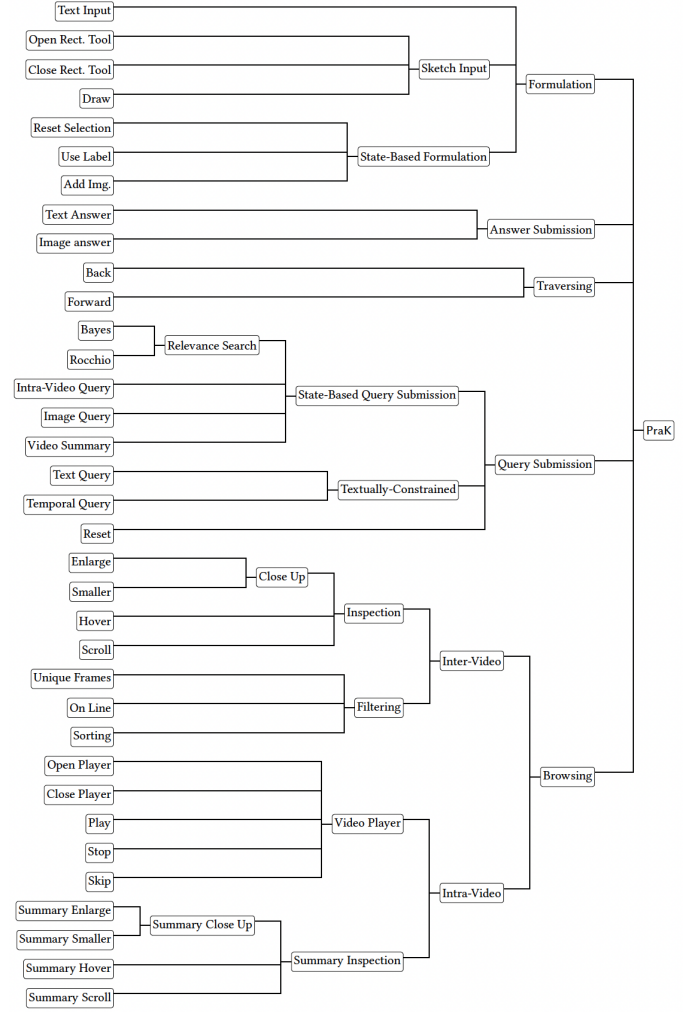


Figure 2: PraK interaction hierarchy

Figure 2. Before we proceed, we first recapitulate the overall results of VBS 2026. In total, 31 tasks were executed across seven task types (described in section 2). The *KIS-V-MVK/KIS-V-LHE* tasks were executed on the MVK [39]/LHE [21] datasets, whereas all other tasks were executed on V3C [28]. For the specific tasks, we refer to the VBS-Repository [23].

Figure 3 summarizes the performance of the analyzed VBS 2026 participants. The heatmap suggests that task characteristics mainly drive both success rates and search times. Of the 31 tasks, 15 were solved by all participants, whereas 2 were solved by none. The between-user differences in aggregate success were limited: top-performing users solved 24 tasks, while the lowest-performing users solved 22. The tasks in the *KIS-V-LHE* and *KIS-C* categories were consistently challenging for the participants. Nevertheless, we also observe user-specific variation in individual tasks. In the following sections, we analyze the collected interaction logs to explain how these shared patterns and user-specific differences emerge from

the search process. We pre-filtered interaction logs to include only those recorded while a task was still active.

4.1 Querying

We start by providing an overview of the most commonly used query actions, both by task and by individual users (see Figure 4). We find that, especially for the V3C collections, text and temporal queries dominate, with only a few cases of relevance updates and image queries. Relevance updates and image queries are most often used for AVS searches and homogeneous datasets. However, text and temporal queries account for approximately 75% of the query types used in these datasets. While the total number of queries strongly depends on the user and systems (with more than 30% differences), it is evident that text queries are used predominantly.

We further analyze which query types lead to success. For KIS tasks, we collect the last query before a successfully submitted answer. For QA tasks, we take the last query before the user found the correct target video (e.g., by loading its video summary or jumping into the video player). We find that in only 13 out of 111 cases, users used either a Relevance Update or an image query. In 11 out of the 13 cases, they were on the homogeneous MVK or LHE datasets. Consequently, finding a target video is an iterative cycle of formulating textual descriptions of scenes, inspecting retrieved candidates, and refining the descriptions.

A major reason users rely heavily on text-only queries is the strong performance of modern multimodal models. Whereas PraK relies on a finetuned MCIP-SigLip variant that improves image-image retrieval [31], Exquisitor relies on standard SigLip [42]. To evaluate retrieval performance, we analyze the rank distributions of target frames and videos for KIS and QA tasks, considering both all issued queries and the final query before a correct answer. We define frame rank as the top-ranked keyframe from the target scene, and video rank as the top-ranked frame from the target video. For QA, we use the same definition, where the target scene is the provided visual reference used to identify the correct video. We report results in Figure 5, where we plot the cumulative distribution function (CDF) of ranks, i.e., for each threshold k , the fraction of queries whose correct item appears at rank $\leq k$. The fraction of top-ranked frames/videos for successful tasks is substantially higher than for unsuccessful tasks. More than 50% of the queries in successful tasks have a target frame rank below 100. If we only consider the last query before a successful answer, the fraction of target frames/videos is even above 80%/90%. We note that only text queries are involved, encompassing around 90% of all queries.

To decouple the influence of the underlying retrieval model from user querying strategies, we performed a cross-system evaluation by re-executing queries from one system on the alternative model (Figure 6). The results reveal a substantial performance gap driven by the source of the query. Queries formulated by PraK users yield higher retrieval rates than those of Exquisitor. Further analysis reveals that PraK users formulated more descriptive queries (mean length 11.34 words, std 6.17) compared to the much shorter inputs in Exquisitor (mean 5.18 words, std 2.40). This suggests that short queries provide insufficient context for effective retrieval.

Regarding retrieval architecture, PraK demonstrates greater robustness, particularly with suboptimal inputs. When processing

Table 1: Comparison of Preprocessing Pipelines

	Exquisitor	PraK
Shot Detection	Provided Default	TransnetV2 [35]
Embedding	SigLIP [42]	MCIP [31]
#Keyframes VBSLHE	75,516	284,410
#Keyframes MVK	34,028	84,309
#Keyframes V3C	4,143,693	4,971,606

the short queries from Exquisitor, the PraK model retrieves targets at consistently better ranks. As the embedding model and keyframe sampling strategy differ between systems (see Table 1), we cannot fully disentangle why Exquisitor shows performance drops. Comparing the SigLip models on real-user queries is future work.

4.2 Browsing

While previous graphs demonstrate the importance of effective text-image search models, browsing is another essential part of interactive search systems. Users must efficiently inspect a vast number of candidates to localize the correct ones. As demonstrated by Joos et al. [10], a good rank of a target is not enough, and there is a non-trivial visual inspection workload to solve a task. Both compared systems present a list of top-ranked candidate frames for KIS tasks, where PraK additionally offers three display types¹.

Figure 7 shows a quantized histogram of the duration of browsing sessions between two reloads of a display in the left part of the figure. The middle and right parts show a quantized histogram of durations between two queries, with the largest value in the bin corresponding to 10-20 seconds between two queries. The adjacent two bins (0-10s and 20-30s) contain slightly lower (but still high) numbers of occurrences, while the remaining bins for longer time between two queries show decreasing numbers in a long tail distribution. The high number of occurrences in the first bin (0-10s) might correspond to fast text query edits/updates or search patterns in which a text query is promptly followed by an image query/relevance search selected from the result set. In contrast, the most frequent durations between display reloads are shorter than 10 seconds, indicating modifications of the display layout during the browsing phase. Note that layout updates comprise everything that triggers a display reload. For PraK, these are all queries (Bayes/Rocchio updates, image query, text query types, intra-video querying, video summary), all layout options (e.g., re-arrangements, unique video frames), reset button, back/forth button. For Exquisitor, these are only the text query, relevance feedback updates, and traversing operations. It is apparent that users are mostly (~ 2/3 of the cases) reformulating their queries no later than 30 seconds after a previous query is issued, and that every other query is likely followed by a layout change interaction for PraK.

The browsing phase can also be analyzed by means of unique frames explored during a task. Figure 8 illustrates the per-category distributions of the unique frames visited by PraK users. Three distributions with the highest mean correspond to AVS, KIS-V-LHE, and KIS-C categories. This result is expected as AVS tasks usually

¹ Exquisitor supports two display types as well, but the video-based frame organization was not used at VBS according to logs.

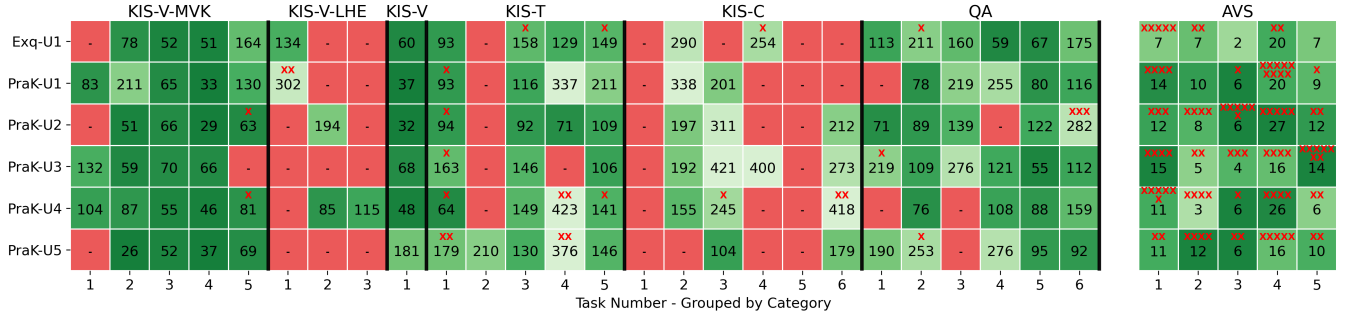


Figure 3: Heatmap of user performances per task. Color is scaled by individual tasks, with darker green indicating better performance, whereas red cells indicate unsolved tasks. Red crosses are incorrect submissions. For AVS, we state the number of correct submissions; for all other tasks, we show the search time in seconds.

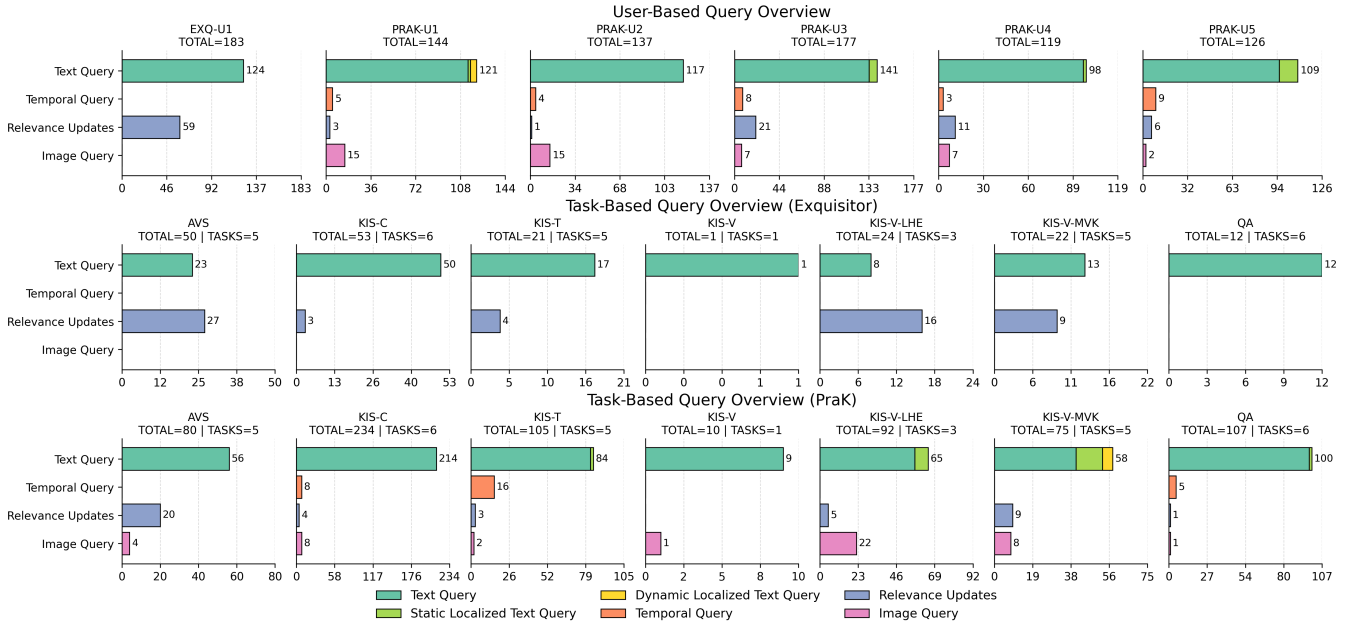
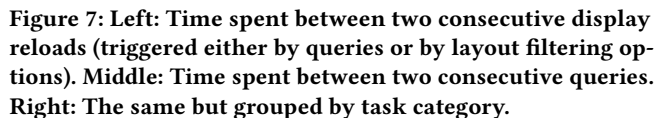
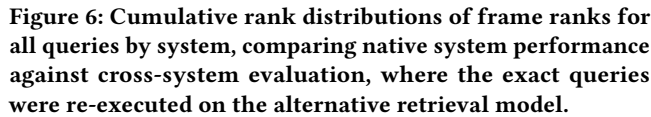
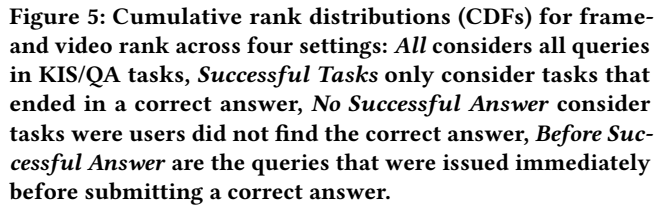


Figure 4: User- and task-based overview of query behavior. The top row shows one bar chart per user; the bottom row shows one bar chart per task category. Colors encode query types (text query, temporal query, relevance updates, and image queries). We further specify whether queries used localization [8] (only implemented for MVK and LHE tasks in PraK).

require exploration of many different videos (more videos can be correct), and KIS-C is a task with an initially unclear goal and potentially changing search intents over time. The KIS-V-LHE category, on the other hand, uses one of the most challenging datasets at VBS, where many visually similar frames from the medical domain need to be explored to identify the one particular moment. On the contrary, the low number of unique frames explored in the KIS-V-V3C category indicates lower difficulty, as users can focus on the rich description of distinct (yet well-known) concepts presented in the video. The KIS-V-MVK category also shows a low number of unique frames explored, which was caused by the simpler tasks selected for this iteration of VBS. Overall, users have to observe thousands of images during the competition, which might introduce non-trivial

effects on the user experience (e.g., fatigue) and affect differences observed between systems.

Observing such a large number of images often leads to overlooking, where the correct target frame is displayed, but not recognized by the user. The exact scrolling data in PraK allows us to estimate the number of overlooks by counting cases where a target frame is on the screen, and subsequently leaves it (without being submitted as an answer). Figure 9 shows the overall number of overlooked target frames for KIS categories for PraK. In total, the KIS-T and KIS-C categories exhibit a much higher number of overlooks compared to all three KIS-V categories. While KIS-V categories are initiated by the exact visual appearance of searched targets, textual KIS tasks rely on subjective user imagination. Even though textual KIS tasks



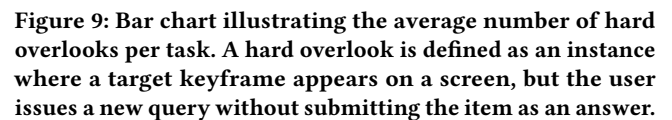
Unique Frames Explored per Task

Count of Unique Items

Legend: Median, IQR (25-75%), Mean, Individual Task

Task	Median	IQR (25-75%)	Mean
KIS-V (WV)	~100	~100 - 220	~150
KIS-T (V2C)	~150	~150 - 320	~250
KIS-V (LHE)	~250	~240 - 540	~400
QA (V2C)	~150	~140 - 300	~250
AVE (V2C)	~250	~220 - 400	~330
KIS-C (V2C)	~280	~220 - 520	~380
KIS-V (V2C)	~50	~40 - 180	~80

Figure 8: Distribution of unique frames explored (derived from PraK logs), categorized by task type and dataset.



Next to inter-video browsing, users spent a substantial amount of time on other interactions. Figure 10 summarizes the estimated time spent in six high-level interaction categories: query formulation, inter-video inspection, intra-video inspection, query submission, answer submission, and traversing. Since the logs record discrete interaction events rather than continuous activity intervals, durations are approximated from consecutive timestamps. Let A and B be two consecutive logged actions with timestamps $t_A < t_B$ and categories c_A and c_B , respectively. We assign the interval $\Delta t = t_B - t_A$ to the category of the later action, c_B . Thus, each bar shows the total estimated time associated with actions of that category. The

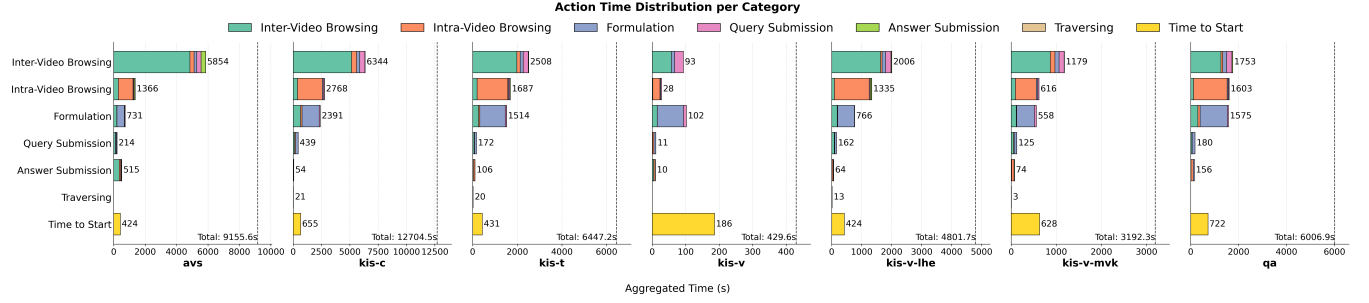


Figure 10: Bar plots of aggregated action times per task category. Time is assigned to the current action as the difference to the previous logged event, considering only events after the initialization phase. Each bar corresponds to the current action, and stacked colored segments indicate the preceding action from which the user transitioned. Time to Start denotes the delay between task start and the first task-relevant action.

stacked colors within a bar indicate the category of the preceding action, c_A , thereby showing from which interaction categories users transitioned before performing actions in the current category. In all task categories, most time was spent during inter-video browsing. Especially, the AVS and KIS-C categories exhibit a high level of these interactions. The intra-video inspection interactions usually reached 10% to 25% of the overall search time, with the highest ratio in the KIS-V-LHE and QA category. The popularity of intra-video inspection interactions indicates the importance of this feature for both systems. It is notable that query formulation also takes a non-trivial amount of time in most categories.

4.3 Discussion & Future Work

Our analysis reveals that, most of the time, the search process alternates between issuing text queries and rapidly viewing large numbers of images. This iterative cycle has a high frequency, with time intervals of less than 30 seconds. The findings highlight the central importance of text-based queries in interactive video retrieval, suggesting that successful search is largely driven by repeated query reformulation combined with fast visual inspection of ranked results. Nevertheless, alternative strategies such as image-based search and relevance feedback remain viable and occasionally lead to correct targets, particularly in homogeneous collections where textual descriptions alone may be insufficient for discrimination. Given the heavy reliance on text queries, future work should investigate query expansion strategies that address both natural-language phrasing mismatches with multimodal text encoders and synonymous expressions of the same concept, thereby increasing the likelihood of retrieving relevant videos and improving robustness across diverse tasks [32, 37]. The motivation behind query reformulation remains underexplored, but with interaction logs, we have the opportunity to examine when reformulations occur, what changes users make, and how viewed results and competition context may shape decisions. Beyond aggregate statistics, fine-grained interaction logging at the lowest level enables search-path-specific analyses and visualizations that can uncover deeper insights into user browsing behavior. However, realizing this potential requires effective visualization techniques and interaction designs that make complex search paths interpretable for both researchers and practitioners.

5 Conclusion

This paper presents the first comprehensive interaction log analysis aiming at explaining the performance of two successful tools participating in the Video Browser Showdown benchmark. To provide such analysis, interaction sets and hierarchies were defined. Furthermore, the logging frameworks were implemented for both systems prior to VBS 2026. Based on the joint terminology derived from the interaction trees, the log analysis presents user- and task-based overviews of the issued queries. As expected, the majority of queries were text-based, demonstrating strong performance in known-item search and question-answering tasks. The times between display reloads and the times to issue the next queries reveal that short delays ($< 30s$) between display changes are most common. Distributions of unique frames inspected during search have the highest mean for tasks with open-ended (usually limited only by the task time limit) or, in general, challenging tasks. The logs also revealed approximate times spent on query formulation and inter-/intra-video browsing, which takes the majority of task-solving time. For PraK, a large number of overlooks of target frames were identified in textual KIS tasks, pointing to open challenges in frame layout design. Overall, the analysis provides a comprehensive explanation of VBS system performance, revealing more than just the overall score and target ranks derived from logged result sets and enabling improvements of the systems.

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References

- [1] Nilavra Bhattacharya and Jacek Gwizdzka. 2021. YASBIL: Yet Another Search Behaviour (and) Interaction Logger. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Virtual Event, Canada) (*SIGIR '21*). Association for Computing Machinery, New York, NY, USA, 2585–2589. doi:10.1145/3404835.3462800
- [2] Jia Chen, Jiaxin Mao, Yiqun Liu, Fan Zhang, Min Zhang, and Shaoping Ma. 2021. Towards a Better Understanding of Query Reformulation Behavior in Web Search. In *Proceedings of the Web Conference 2021* (Ljubljana, Slovenia) (*WWW '21*). Association for Computing Machinery, New York, NY, USA, 743–755. doi:10.1145/3442381.3450127
- [3] Michael J. Cole, Chathra Hendahewa, Nicholas J. Belkin, and Chirag Shah. 2015. User Activity Patterns During Information Search. *ACM Trans. Inf. Syst.* 33, 1, Article 1 (March 2015), 39 pages. doi:10.1145/2699656
- [4] Silvan Heller, Viktor Gsteiger, Werner Bailer, Cathal Gurrin, Björn Þór Jónsson, Jakub Lokoč, Andreas Leibetseder, František Mejzlík, Ladislav Peška, Luca Rossetto, Konstantin Schall, Klaus Schoeffmann, Heiko Schuldt, Florian Spiess, Ly-Duyen Tran, Lucia Vadicamo, Patrik Veselý, Stefanos Vrochidis, and Jiaxin Wu. 2022. Interactive video retrieval evaluation at a distance: comparing sixteen interactive video search systems in a remote setting at the 10th Video Browser Showdown. *International Journal of Multimedia Information Retrieval* 11, 1 (2022), 1–18. doi:10.1007/s13735-021-00225-2
- [5] Daniel Hienert and Peter Mutschke. 2016. A Usefulness-based Approach for Measuring the Local and Global Effect of IIR Services. In *Proceedings of the 2016 ACM Conference on Human Information Interaction and Retrieval* (Carrboro, North Carolina, USA) (*CHIIR '16*). Association for Computing Machinery, New York, NY, USA, 153–162. doi:10.1145/2854946.2854962
- [6] Jeff Huang and Efthimis N. Efthimiadis. 2009. Analyzing and evaluating query reformulation strategies in web search logs. In *Proceedings of the 18th ACM Conference on Information and Knowledge Management* (Hong Kong, China) (*CIKM '09*). Association for Computing Machinery, New York, NY, USA, 77–86. doi:10.1145/1645953.1645966
- [7] ICMR2025 2025. *ICMR '25: Proceedings of the 2025 International Conference on Multimedia Retrieval* (Chicago, IL, USA). Association for Computing Machinery, New York, NY, USA.
- [8] Bastian Jäckl, Vojtěch Kloda, Daniel A. Keim, and Jakub Lokoč. 2026. Dynamic Sub-region Search In Homogeneous Collections Using CLIP. In *Similarity Search and Applications*. Springer Nature Switzerland, Cham, 105–118. doi:10.1007/978-3-032-06069-3_9
- [9] Jiepu Jiang, Daqing He, and James Allan. 2014. Searching, browsing, and clicking in a search session: changes in user behavior by task and over time. In *Proceedings of the 37th International ACM SIGIR Conference on Research & Development in Information Retrieval* (Gold Coast, Queensland, Australia) (*SIGIR '14*). Association for Computing Machinery, New York, NY, USA, 607–616. doi:10.1145/2600428.2609633
- [10] Lucas Joos, Bastian Jäckl, Daniel A. Keim, Maximilian T. Fischer, Ladislav Peška, and Jakub Lokoč. 2024. Known-Item Search in Video: An Eye Tracking-Based Study. In *Proceedings of the 2024 International Conference on Multimedia Retrieval, ICMR 2024, Phuket, Thailand, June 10-14, 2024*, Cathal Gurrin, Rachada Kongkachandra, Klaus Schoeffmann, Duc-Tien Dang-Nguyen, Luca Rossetto, Shin'ichi Satoh, and Liting Zhou (Eds.). ACM, New York, NY, USA, 311–319. doi:10.1145/3652583.3658119
- [11] Yidong Liang, Zhijiang Wu, Fan Zhang, Dandan Song, and Heyan Huang. 2025. How Users Interact with Generative Information Retrieval Systems: A Study of User Behavior and Search Experience. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Padua, Italy) (*SIGIR '25*). Association for Computing Machinery, New York, NY, USA, 634–644. doi:10.1145/3726302.3729998
- [12] Jakub Lokoč, Stelios Andreadis, Werner Bailer, Aaron Duane, Cathal Gurrin, Zhixin Ma, Nicola Messina, Thao-Nhu Nguyen, Ladislav Peška, Luca Rossetto, Loris Sauter, Konstantin Schall, Klaus Schoeffmann, Omar Shahbaz Khan, Florian Spiess, Lucia Vadicamo, and Stefanos Vrochidis. 2023. Interactive video retrieval in the age of effective joint embedding deep models: lessons from the 11th VBS. *Multimedia Syst.* 29, 6 (Aug. 2023), 3481–3504. doi:10.1007/s00530-023-01143-5
- [13] Jakub Lokoč, Gregor Kovalčík, Bernd Münzer, Klaus Schöffmann, Werner Bailer, Ralph Gasser, Stefanos Vrochidis, Phuong Anh Nguyen, Sitapa Rujikietgumjorn, and Kai Uwe Barthel. 2019. Interactive Search or Sequential Browsing? A Detailed Analysis of the Video Browser Showdown 2018. *ACM Trans. Multimedia Comput. Commun. Appl.* 15, 1, Article 29 (Feb. 2019), 18 pages. doi:10.1145/3295663
- [14] Jakub Lokoč, Gregor Kovalčík, Tomáš Souček, Jaroslav Moravec, and Přemysl Čech. 2019. A Framework for Effective Known-item Search in Video. In *Proceedings of the 27th ACM International Conference on Multimedia*, MM 2019, Nice, France, October 21–25, 2019, Laurent Amsaleg, Benoit Huet, Martha A. Larson, Guillaume Gravier, Hayley Hung, Chong-Wah Ngo, and Wei Tsang Ooi (Eds.). ACM, New York, NY, USA, 1777–1785. doi:10.1145/3343031.3351046
- [15] Jakub Lokoč, Gregor Kovalčík, Tomáš Souček, Jaroslav Moravec, and Přemysl Čech. 2019. VIRET: A Video Retrieval Tool for Interactive Known-item Search. In *Proceedings of the 2019 on International Conference on Multimedia Retrieval, ICMR 2019, Ottawa, ON, Canada, June 10-13, 2019*, Abdulmotaleb El-Saddik, Alberto Del Bimbo, Zhongfei Zhang, Alexander G. Hauptmann, K. Selçuk Candan, Marco Bertini, Lexing Xie, and Xiao-Yong Wei (Eds.). ACM, New York, NY, USA, 177–181. doi:10.1145/3323873.3325034
- [16] Jakub Lokoč, Tomáš Souček, Patrik Veselý, František Mejzlík, Jiaqi Ji, Chaoxi Xu, and Xirong Li. 2020. A W2VV++ Case Study with Automated and Interactive Text-to-Video Retrieval. In *Proceedings of the 28th ACM International Conference on Multimedia* (Seattle, WA, USA) (*MM '20*). Association for Computing Machinery, New York, NY, USA, 2553–2561. doi:10.1145/3394171.3414002
- [17] Jakub Lokoč, Patrik Veselý, František Mejzlík, Gregor Kovalčík, Tomáš Souček, Luca Rossetto, Klaus Schoeffmann, Werner Bailer, Cathal Gurrin, Loris Sauter, Jaeyub Song, Stefanos Vrochidis, Jiaxin Wu, and Björn Þór Jónsson. 2020. Is the Reign of Interactive Search Eternal? Findings from the Video Browser Showdown 2020. *ACM Trans. Multimedia Comput. Commun. Appl.* 17, 3, Article 91 (July 2021), 26 pages. doi:10.1145/3445031
- [18] Jakub Lokoč, Zuzana Vopálková, Michael Stroh, Raphael Buchmüller, and Udo Schlegel. 2024. PraK Tool: An Interactive Search Tool Based on Video Data Services. In *MultiMedia Modeling*, Stevan Rudinac, Alan Hanjalic, Cynthia Liem, Marcel Worring, Björn Þór Jónsson, Bei Liu, and Yoko Yamakata (Eds.). Springer Nature Switzerland, Cham, 340–346. doi:10.1007/978-3-031-53302-0_30
- [19] Wanxuan Lu and Yunde Jia. 2014. An Eye-Tracking Study of User Behavior in Web Image Search. In *PRICAI 2014: Trends in Artificial Intelligence*, Duc-Nghia Pham and Seong-Bae Park (Eds.). Springer International Publishing, Cham, 170–182. doi:10.1007/978-3-319-13560-1_14
- [20] MMM2025 2025. *MultiMedia Modeling: 31st International Conference on Multimedia Modeling*, MMM 2025, Nara, Japan, January 8–10, 2025, *Proceedings, Part II*. Springer.
- [21] Sahar Nasirihaghighi, Negin Ghamsarian, Leonie Peschek, Matteo Munari, Heinrich Husslein, Raphael Sznitman, and Klaus Schoeffmann. 2025. GynSurG: A Comprehensive Gynecology Laparoscopic Surgery Dataset. In *Proceedings of the 33rd ACM International Conference on Multimedia* (Dublin, Ireland) (*MM '25*). Association for Computing Machinery, New York, NY, USA, 13141–13147. doi:10.1145/3746027.3758267
- [22] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*. PmLR, PMLR, Virtual, 8748–8763.
- [23] Luca Rossetto. 2026. VBS-Archive. <https://github.com/lucaro/VBS-Archive>. GitHub repository: Archive of Tasks and Results of the Video Browser Showdown. Accessed: 2026-04-22.
- [24] Luca Rossetto, Werner Bailer, Cathal Gurrin, Duc Tien Dang Nguyen, Klaus Schoeffmann, and Allie Tran. 2025. Overview of the First CASTLE Grand Challenge at ACM Multimedia 2025. In *Proceedings of the 33rd ACM International Conference on Multimedia* (Dublin, Ireland) (*MM '25*). Association for Computing Machinery, New York, NY, USA, 14271–14272. doi:10.1145/3746027.3760242
- [25] Luca Rossetto, Ralph Gasser, Silvan Heller, Mahnaz Parian-Scherb, Loris Sauter, Florian Spiess, Heiko Schuldt, Ladislav Peška, Tomáš Souček, Miroslav Kratochvíl, František Mejzlík, Patrik Veselý, and Jakub Lokoč. 2021. On the User-Centric Comparative Remote Evaluation of Interactive Video Search Systems. *IEEE MultiMedia* 28, 4 (2021), 18–28. doi:10.1109/MMUL.2021.3066779
- [26] Luca Rossetto, Ralph Gasser, Jakub Lokoč, Werner Bailer, Klaus Schoeffmann, Bernd Muenzer, Tomáš Souček, Phuong Anh Nguyen, Paolo Bolettieri, Andreas Leibetseder, and Stefanos Vrochidis. 2021. Interactive Video Retrieval in the Age of Deep Learning – Detailed Evaluation of VBS 2019. *IEEE Transactions on Multimedia* 23 (2021), 243–256.

- [27] Luca Rossetto, Ralph Gasser, Loris Sauter, Abraham Bernstein, and Heiko Schuldt. 2021. A System for Interactive Multimedia Retrieval Evaluations. In *MultiMedia Modeling*, Jakub Lokoč, Tomáš Skopal, Klaus Schoeffmann, Vasileios Mezaris, Xirong Li, Stefanos Vrochidis, and Ioannis Patras (Eds.). Springer International Publishing, Cham, 385–390. doi:10.1007/978-3-030-67835-7_33
- [28] Luca Rossetto, Heiko Schuldt, George Awad, and Asad A. Butt. 2019. V3C - A Research Video Collection. In *MMM*. Springer International Publishing, Cham, 349–360. doi:10.1007/978-3-030-05710-7_29
- [29] Loris Sauter, Ralph Gasser, Heiko Schuldt, Abraham Bernstein, and Luca Rossetto. 2024. Performance Evaluation in Multimedia Retrieval. *ACM Transactions on Multimedia Computing, Communications and Applications* 21 (Oct. 2024), 23 pages. doi:10.1145/3678881
- [30] Konstantin Schall, Werner Bailer, Kai Barthel, Fabio Carrara, Jakub Lokoc, Ladislav Peska, Klaus Schoeffmann, Lucia Vadicamo, and Claudio Vairo. 2024. Interactive multimodal video search: an extended post-evaluation for the VBS 2022 competition. *International Journal of Multimedia Information Retrieval* 13 (2024), 13 pages. doi:10.1007/s13735-024-00325-9
- [31] Konstantin Schall, Kai Uwe Barthel, Nico Hezel, and Klaus Jung. 2024. Optimizing CLIP Models for Image Retrieval with Maintained Joint-Embedding Alignment. In *Similarity Search and Applications: 17th International Conference, SISAP 2024, Providence, RI, USA, November 4–6, 2024, Proceedings* (Providence, RI, USA). Springer-Verlag, Berlin, Heidelberg, 97–110. doi:10.1007/978-3-031-75823-2_9
- [32] Udo Schlegel, Franziska Weeber, Jian Lan, and Thomas Seidl. 2026. PRSM: A Measure to Evaluate CLIP's Robustness Against Paraphrases. In *International Conference on Multimedia Modeling*. Springer, Springer-Verlag, Berlin, Heidelberg, 72–79. doi:10.1007/978-981-95-6963-2_6
- [33] Klaus Schoeffmann, Marco A. Hudelist, and Jochen Huber. 2015. Video Interaction Tools: A Survey of Recent Work. *ACM Comput. Surv.* 48, 1, Article 14 (Sept. 2015), 34 pages. doi:10.1145/2808796
- [34] Ujjwal Sharma, Omar Shahbaz Khan, Stevan Rudinac, and Björn Þór Jónsson. 2025. Exquisitor at the Video Browser Showdown 2025: Unifying Conversational Search and User Relevance Feedback. In *MultiMedia Modeling: 31st International Conference on Multimedia Modeling, MMM 2025, Nara, Japan, January 8–10, 2025, Proceedings, Part V* (Nara, Japan). Springer-Verlag, Berlin, Heidelberg, 264–271. doi:10.1007/978-981-96-2074-6_31
- [35] Tomáš Souček and Jakub Lokoc. 2019. TransNet V2: An Effective Deep Network for Fast Shot Transition Detection. In *MM '24: Proceedings of the 32nd ACM International Conference on Multimedia*. ACM, New York, NY, USA, 11218–1122. doi:10.1145/3664647.3685517
- [36] Allie Tran, Werner Bailer, Duc-Tien Dang-Nguyen, Graham Healy, Steve Hodges, Björn Þór Jónsson, Luca Rossetto, Klaus Schöffmann, Minh-Triet Tran, Lucia Vadicamo, and Cathal Gurrin. 2025. The State-of-the-Art in Lifelog Retrieval: A Review of Progress at the ACM Lifelog Search Challenge Workshop 2022-24. *ArXiv abs/2506.06743* (2025), 24 pages. doi:10.1109/ACCESS.2025.3644952
- [37] Allie Tran and Luca Rossetto. 2026. On the Brittleness of CLIP Text Encoders. In *MultiMedia Modeling: 32nd International Conference on Multimedia Modeling, MMM 2026, Prague, Czech Republic, January 29–31, 2026, Proceedings, Part II* (Prague, Czech Republic). Springer-Verlag, Berlin, Heidelberg, 361–374. doi:10.1007/978-981-95-6957-1_26
- [38] Ly-Duyen Tran, Manh-Duy Nguyen, Duc-Tien Dang-Nguyen, Silvan Heller, Florian Spiess, Jakub Lokoč, Ladislav Peška, Thao-Nhu Nguyen, Omar Shahbaz Khan, Aaron Duane, Björn Þór Jónsson, Luca Rossetto, An-Zi Yen, Ahmed Alateeq, Naushad Alam, Minh-Triet Tran, Graham Healy, Klaus Schoeffmann, and Cathal Gurrin. 2023. Comparing Interactive Retrieval Approaches at the Lifelog Search Challenge 2021. *IEEE Access* 11 (2023), 30982–30995. doi:10.1109/ACCESS.2023.3248284
- [39] Quang-Trung Truong and et al. 2023. Marine Video Kit: A New Marine Video Dataset for Content-Based Analysis and Retrieval. In *MMM*, Vol. 13833. Springer, Berlin, Heidelberg, 539–550. doi:10.1007/978-3-031-27077-2_42
- [40] Lucia Vadicamo, Rahel Arnold, Werner Bailer, Fabio Carrara, Cathal Gurrin, Nico Hezel, Xinghan Li, Jakub Lokoc, Sebastian Lubos, Zhixin Ma, Nicola Messina, Thao-Nhu Nguyen, Ladislav Peska, Luca Rossetto, Loris Sauter, Klaus Schöffmann, Florian Spiess, Minh-Triet Tran, and Stefanos Vrochidis. 2024. Evaluating Performance and Trends in Interactive Video Retrieval: Insights From the 12th VBS Competition. *IEEE Access* 12 (2024), 79342–79366. doi:10.1109/ACCESS.2024.3405638
- [41] Orion Weller, Michael Boratko, Iftekhar Naim, and Jinhyuk Lee. 2026. On the Theoretical Limitations of Embedding-Based Retrieval. In *The Fourteenth International Conference on Learning Representations*.
- [42] Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. 2023. Sigmoid loss for language image pre-training. In *Proceedings of the IEEE/CVF international conference on computer vision*. IEEE, Paris, France, 11975–11986. doi:10.1109/ICCV51070.2023.01100.